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Camera optical lens

Abstract

The present invention discloses a camera optical lens, including from an object side to an image side in sequence: a first lens having a negative refractive power, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens; wherein the camera optical lens satisfies the following conditions: $100.00^\circ \leq \text{FOV} \leq 135.00^\circ$, $-3.20 \leq f1/f \leq -1.00$, and $2.30 \leq R15/R16 \leq 5.00$. FOV denotes a field of view of the camera optical lens in a diagonal direction. F denotes a focal length of the camera optical lens. f1 denotes a focal length of the first lens. R15 denotes a central curvature radius of an object side surface of the eighth lens. R16 denotes a central curvature radius of an image side surface of the eighth lens. The camera optical lens according to the present invention has a wide-angle effect.

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Background/Summary

FIELD OF THE PRESENT INVENTION

(1) The present invention relates to the field of optical lens, and more particularly, to a camera optical lens suitable for handheld terminal devices, such as smart phones and digital cameras, monitors or PC lenses.

DESCRIPTION OF RELATED ART

(2) In recent years, with the rise of various smart devices, the demand for miniaturized camera optics has been increasing, and the pixel size of photosensitive devices has shrunk, coupled with the development trend of electronic products with good functions, thin and portable appearance, Therefore, miniaturized imaging optical lenses with good image quality have become the mainstream in the current market. In order to obtain better imaging quality, a multi-piece lenses structure is often used. Moreover, with the development of technology and the increase of diversified needs of users, as the pixel area of the photosensitive device continues to shrink and the system's requirements for image quality continue to increase, the eight-piece lenses structure gradually appears in the lens design. There is an urgent need for a wide-angle imaging lens with excellent optical characteristics, small size, and fully corrected aberrations.

SUMMARY

(3) In the present invention, a camera optical lens has a wide-angle effect.

(4) According to one aspect of the present invention, a camera optical lens comprises, from an object side to an image side in sequence, a first lens having a negative refractive power, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens, wherein the camera optical lens satisfies the following conditions: $100.00^\circ \leq \text{FOV} \leq 135.00^\circ$, $-3.20 \leq f_1/f \leq -1.00$, and $2.30 \leq R_{15}/R_{16} \leq 5.00$. FOV denotes a field of view in a diagonal direction of the camera optical lens. f denotes a focal length of the camera optical lens, f₁ denotes a focal length of the first lens, R₁₅ denotes a central curvature radius of an object side surface of the eighth lens, and R₁₆ denotes a central curvature radius of an image side surface of the eighth lens.

(5) As an improvement, the fifth lens has a negative refractive power. The camera optical lens satisfies the following condition: $-4.50 \leq f_5/f \leq -2.00$. f₅ denotes a focal length of the fifth lens.

(6) As an improvement, the first lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $0.15 \leq (R_1 + R_2)/(R_1 - R_2) \leq 0.88$ and $0.03 \leq d_1/\text{TTL} \leq 0.20$. R₁ denotes a central curvature radius of the object side surface of the first lens, R₂ denotes a central curvature radius of the image side surface of the first lens, d₁ denotes an on-axis thickness of the first lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

(7) As an improvement, the camera optical lens further satisfies the following conditions: $0.23 \leq (R_1 + R_2)/(R_1 - R_2) \leq 0.71$ and $0.04 \leq d_1/\text{TTL} \leq 0.16$.

(8) As an improvement, the second lens has an object side surface being convex in a paraxial region, an image side surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $-17.11 \leq f_2/f \leq 4.12$, $-7.01 \leq (R_3 + R_4)/(R_3 - R_4) \leq 6.85$, and $0.01 \leq d_3/\text{TTL} \leq 0.05$. f₂ denotes a focal length of the second lens, R₃ denotes a central curvature radius of the object side surface of the second lens, R₄ denotes a central curvature radius of the image side surface of the second lens, d₃ denotes an on-axis thickness of the second lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

(9) As an improvement, the camera optical lens further satisfies the following conditions: $-10.69 \leq f_2/f \leq 3.30$, $-4.38 \leq (R_3 + R_4)/(R_3 - R_4) \leq 5.48$, and $0.01 \leq d_3/\text{TTL} \leq 0.04$.

(10) As an improvement, the third lens has an object side surface being convex in a paraxial region, an image side surface being concave in the paraxial region, and the third lens has a positive refractive power. The camera optical lens further satisfies the following conditions: $1.21 \leq f_3/f \leq 408.22$, $-20.33 \leq (R_5 + R_6)/(R_5 - R_6) \leq -1.27$, and $0.02 \leq d_5/\text{TTL} \leq 0.09$. f₃ denotes a focal length of the third lens, R₅ denotes a central curvature radius of the object side surface of the third lens, R₆ denotes a central curvature radius of the image side surface of the third lens, d₅ denotes an on-axis thickness of the third lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

(11) As an improvement, the camera optical lens further satisfies the following conditions: $1.93 \leq f_3/f \leq 326.58$, $-12.70 \leq (R_5 + R_6)/(R_5 - R_6) \leq -1.58$, and $0.03 \leq d_5/\text{TTL} \leq 0.07$.

(12) As an improvement, the fourth lens has an object side surface being convex in a paraxial region, an image side surface being convex in the paraxial region, and the fourth lens has a positive refractive power. The camera optical lens further satisfies the following conditions: $0.59 \leq f_4/f \leq 2.14$, $0.07 \leq (R_7 + R_8)/(R_7 - R_8) \leq 0.46$, and $0.04 \leq d_7/\text{TTL} \leq 0.18$. f₄ denotes a focal length of the fourth lens, R₇ denotes a central curvature radius of the object side surface of the fourth lens, R₈ denotes a central curvature radius of the image side surface of the fourth lens, d₇ denotes an on-axis thickness of the fourth lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

(13) As an improvement, the camera optical lens further satisfies the following conditions: $0.95 \leq f_4/f \leq 1.71$, $0.11 \leq (R_7 + R_8)/(R_7 - R_8) \leq 0.37$, and $0.07 \leq d_7/\text{TTL} \leq 0.14$.

(14) As an improvement, the fifth lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $0.37 \leq (R_9 + R_{10})/(R_9 - R_{10}) \leq 1.42$ and $0.01 \leq d_9/\text{TTL} \leq 0.08$. R₉ denotes a central curvature radius of the object side surface of the fifth lens, R₁₀ denotes a central curvature radius of the image side surface of the fifth lens, d₉ denotes an on-axis thickness of the fifth lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

- (15) As an improvement, the camera optical lens further satisfies the following conditions: $0.60 \leq (R9+R10)/(R9-R10) \leq 1.14$ and $0.02 \leq d9/TTL \leq 0.07$.
- (16) As an improvement, the camera optical lens further satisfies the following conditions: $-645.45 \leq f_6/f \leq 138.30$, $0.35 \leq (R11+R12)/(R11-R12) \leq 64.63$, and $0.01 \leq d11/TTL \leq 0.12$. f_6 denotes a focal length of the sixth lens, $R11$ denotes a central curvature radius of an object side surface of the sixth lens, $R12$ denotes a central curvature radius of an image side surface of the sixth lens, $d11$ denotes an on-axis thickness of the sixth lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
- (17) As an improvement, the camera optical lens further satisfies the following conditions: $-403.41 \leq f_6/f \leq 110.64$, $0.57 \leq (R11+R12)/(R11-R12) \leq 51.70$, and $0.01 \leq d11/TTL \leq 0.09$.
- (18) As an improvement, the seventh lens has an object side surface being convex in a paraxial region and an image side surface being convex in a paraxial region, and the seventh lens has a positive refractive power. The camera optical lens further satisfies the following conditions: $0.40 \leq f_7/f \leq 1.59$, $0.38 \leq (R13+R14)/(R13-R14) \leq 1.29$, and $0.05 \leq d13/TTL \leq 0.31$. f_7 denotes a focal length of the seventh lens, $R13$ denotes a central curvature radius of the object side surface of the seventh lens, $R14$ denotes a central curvature radius of the image side surface of the seventh lens, $d13$ denotes an on-axis thickness of the seventh lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
- (19) As an improvement, the camera optical lens further satisfies the following conditions: $0.64 \leq f_7/f \leq 1.27$, $0.60 \leq (R13+R14)/(R13-R14) \leq 1.03$, and $0.08 \leq d13/TTL \leq 0.25$.
- (20) As an improvement, the object side surface of the eighth lens is convex in a paraxial region, the image side surface of the eighth lens is concave in the paraxial region, and the eighth lens has a negative refractive power. The camera optical lens further satisfies the following conditions: $-2.96 \leq f_8/f \leq -0.69$, $0.75 \leq (R15+R16)/(R15-R16) \leq 3.80$, and $0.03 \leq d15/TTL \leq 0.15$. f_8 denotes a focal length of the eighth lens, $d15$ denotes an on-axis thickness of the eighth lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
- (21) As an improvement, the camera optical lens further satisfies the following conditions: $-1.85 \leq f_8/f \leq -0.86$, $1.20 \leq (R15+R16)/(R15-R16) \leq 3.04$, and $0.05 \leq d15/TTL \leq 0.12$.
- (22) As an improvement, the camera optical lens further satisfies the following condition: $TTL/IH \leq 2.91$. IH denotes an image height of the camera optical lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.
- (23) As an improvement, an FNO of the camera optical lens is less than or equal to 2.68. FNO denotes a ratio of an effective focal length of the camera optical lens to an entrance pupil diameter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In order to explain the technical solutions in the embodiments of the present invention more clearly, the following will briefly introduce the drawings that need to be used in the description of the embodiments. Obviously, the drawings in the following description are only some embodiments of the present invention. For those of ordinary skill in the art, without creative work, other drawings can be obtained based on these drawings, among which:
- (2) FIG. 1 is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 1 of the present invention;
- (3) FIG. 2 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 1;
- (4) FIG. 3 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 1;
- (5) FIG. 4 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 1;
- (6) FIG. 5 is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 2 of the present invention;
- (7) FIG. 6 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 5;
- (8) FIG. 7 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 5;
- (9) FIG. 8 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 5;
- (10) FIG. 9 is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 3 of the present invention;
- (11) FIG. 10 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 9;
- (12) FIG. 11 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 9; and
- (13) FIG. 12 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 9.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(14) In order to make the objects, technical solutions, and advantages of the present invention more apparent, the embodiments of the present invention will be described in detail below. However, it will be apparent to the one skilled in the art that, in the various embodiments of the present invention, a number of technical details are presented in order to provide the reader with a better understanding of the invention. However, the technical solutions claimed in the present invention can be implemented without these technical details and various changes and modifications based on the following embodiments.

Embodiment 1

(15) As referring to the accompanying drawings, the present invention provides a camera optical lens 10. FIG. 1 shows the camera optical lens 10 according to embodiment 1 of the present invention. The camera optical lens 10 comprises eight-piece lenses. Specifically, from an object side to an image side, the camera optical lens 10 comprises in sequence: a first lens L1, a second lens L2, a third lens L3, an aperture S1, a fourth lens L4, a fifth lens L5, a sixth lens L6, a seventh lens L7 and an eighth lens L8. Optical elements like optical filter GF can be arranged between the eighth lens L8 and an image surface S1.

(16) The first lens L1 is made of plastic material, the second lens L2 is made of plastic material, the third lens L3 is made of plastic material, the fourth lens L4 is made of plastic material, the fifth lens L5 is made of plastic material, the sixth lens L6 is made of plastic material, the seventh lens L7 is made of plastic material, and the eighth lens L8 is made of plastic material.

(17) In other optional embodiments, each lens may also be made of other materials, such as glasses.

(18) In the present embodiment, a field of view of the camera optical lens 10 in a diagonal direction is defined as FOV, and the camera optical lens 10 satisfies the following condition: $100.00^\circ \leq \text{FOV} \leq 135.00^\circ$. The range of the FOV is specified. When the camera optical lens 10 satisfies the above condition, the camera optical lens 10 has a wide-angle effect.

(19) A focal length of the camera optical lens 10 is defined as f , and a focal length of the first lens L1 is defined as f_1 . The camera optical

lens 10 meets the following condition: $-3.20 \leq f_1/f \leq -1.00$. A ratio of the focal length f_1 of the first lens L1 to a focal length f of the camera optical lens 10 is specified. When the above condition is satisfied, the performance of the camera optical lens 10 would be improved.

(20) A central curvature radius of an object side surface of the eighth lens L8 is defined as R15, and a central curvature radius of an image side surface of the eighth lens L8 is defined as R16. The camera optical lens 10 meets the following condition: $2.30 \leq R15/R16 \leq 5.00$. The condition specifies a shape of the eighth lens L8. When the condition is satisfied, a light deflection can be alleviated while the light entry into the lens, and an aberration is effectively reduced.

(21) The focal length of the camera optical lens 10 is defined as f , and a focal length of the fifth lens L5 is defined as f_5 . The camera optical lens 10 meets the following condition: $-4.50 \leq f_5/f \leq -2.00$. A ratio of the focal length f_5 of the fifth lens L5 to the focal length f of the camera optical lens is specified in the condition. The camera optical lens 10 has a higher imaging quality and a lower sensitivity as the focal length reasonably distributed. Preferably, the camera optical lens 10 further meets the following condition: $-4.38 \leq f_5/f \leq -2.19$.

(22) In the present embodiment, an object side surface of the first lens L1 is concave in a paraxial region, an image side surface of the first lens L1 is concave in the paraxial region, and the first lens L1 has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the first lens L1 can also be arranged as other concave side surface or convex side surface, such as, convex object side surface and convex image side surface and so on.

(23) A central curvature radius of the object side surface of the first lens L1 is defined as R1, and a central curvature radius of the image side surface of the first lens L1 is defined as R2. The camera optical lens 10 further satisfies the following condition: $0.15 \leq (R1+R2)/(R1-R2) \leq 0.88$. This condition reasonably controls a shape of the first lens L1, so that the first lens L1 can effectively correct a spherical aberration of the camera optical lens 10. Preferably, the following condition shall be satisfied, $0.23 \leq (R1+R2)/(R1-R2) \leq 0.71$.

(24) An on-axis thickness of the first lens L1 is defined as d_1 . A total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along an optical axis is defined as TTL. The camera optical lens 10 further satisfies the following condition: $0.03 \leq d_1/TTL \leq 0.20$. When the value is within this range, it is beneficial for producing an ultra-thin effect. Preferably, the following condition shall be satisfied, $0.04 \leq d_1/TTL \leq 0.16$.

(25) In the present embodiment, an object side surface of the second lens L2 is convex in the paraxial region, the image side surface of the second lens L2 is concave in the paraxial region, and the second lens L2 has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the second lens L2 can also be arranged as other concave side surface or convex side surface, such as, concave object side surface and convex image side surface and so on.

(26) The focal length of the camera optical lens 10 is defined as f , and a focal length of the second lens L2 is defined as f_2 . The camera optical lens 10 further satisfies the following condition: $-17.11 \leq f_2/f \leq 4.12$. It is beneficial for correcting the aberration of the camera optical lens 10 by controlling the positive refractive power of the second lens L2 being within reasonable range. Preferably, the following condition shall be satisfied, $-10.69 \leq f_2/f \leq 3.30$.

(27) A central curvature radius of the object side surface of the second lens L2 is defined as R3, and a central curvature radius of the image side surface of the second lens L2 is defined as R4. The camera optical lens 10 further satisfies the following condition: $-7.01 \leq (R3+R4)/(R3-R4) \leq 6.85$, which specifies a shape of the second lens L2. When the condition is satisfied, as the camera optical lens 10 develops toward ultra-thin and wide-angle lenses, it is beneficial for correcting an on-axis chromatic aberration. Preferably, the following condition shall be satisfied, $-4.38 \leq (R3+R4)/(R3-R4) \leq 5.48$.

(28) An on-axis thickness of the second lens L2 is defined as d_3 . The total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along the optical axis is defined as TTL. The camera optical lens 10 further satisfies the following condition: $0.01 \leq d_3/TTL \leq 0.05$. When the value is within this range, it is beneficial for producing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.01 \leq d_3/TTL \leq 0.04$.

(29) In the present embodiment, an object side surface of the third lens L3 is convex in the paraxial region, an image side surface of the third lens L3 is concave in the paraxial region, and the third lens L3 has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the third lens L3 can also be arranged as other concave side surface or convex side surface, such as, concave object side surface and convex image side surface and so on.

(30) The focal length of the camera optical lens 10 is defined as f , and a focal length of the third lens L3 is defined as f_3 . The camera optical lens 10 further satisfies the following condition: $1.21 \leq f_3/f \leq 408.22$. By a reasonable distribution of the refractive power, which makes the camera optical lens 10 has the higher imaging quality and the lower sensitivity. Preferably, the following condition shall be satisfied, $1.93 \leq f_3/f \leq 326.58$.

(31) A central curvature radius of the object side surface of the third lens L3 is defined as R5, and a central curvature radius of the image side surface of the third lens L3 is defined as R6. The camera optical lens 10 further satisfies the following condition: $-20.33 \leq (R5+R6)/(R5-R6) \leq -1.27$, which specifies a shape of the third lens L3. It is beneficial for molding the third lens L3. When the condition is satisfied, a light deflection can be alleviated while the light passing through the lens, and the aberration can be reduced effectively. Preferably, the following condition shall be satisfied, $-12.70 \leq (R5+R6)/(R5-R6) \leq -1.58$.

(32) An on-axis thickness of the third lens L3 is defined as d_5 . The total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along the optical axis is defined as TTL. The camera optical lens 10 further satisfies the following condition: $0.02 \leq d_5/TTL \leq 0.09$, which benefits for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.03 \leq d_5/TTL \leq 0.07$.

(33) In the present embodiment, the object side surface of the fourth lens L4 is convex in the paraxial region, an image side surface of the fourth lens L4 is convex in the paraxial region, and the fourth lens L4 has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the fourth lens L4 can also be arranged as other convex side surface or concave side surface, such as, concave object side surface and concave image side surface and so on.

(34) The focal length of the camera optical lens 10 is defined as f , and a focal length of the fourth lens L4 is defined as f_4 . The camera optical lens 10 further satisfies the following condition: $0.59 \leq f_4/f \leq 2.14$. It is beneficial for realizing the excellent imaging quality and the lower sensitivity by controlling the refractive power being within reasonable range. Preferably, the following condition shall be satisfied, $0.95 \leq f_4/f \leq 1.71$.

(35) A curvature radius of the object side surface of the fourth lens L4 is defined as R7, and a central curvature radius of the image side surface of the fourth lens L4 is defined as R8. The camera optical lens further satisfies the following condition: $0.07 \leq (R7+R8)/(R7-R8) \leq 0.46$, which specifies a shape of the fourth lens L4. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for solving the problems, such as correcting an off-axis aberration. Preferably, the

following condition shall be satisfied, $0.11 \leq (R7+R8)/(R7-R8) \leq 0.37$.

(36) An on-axis thickness of the fourth lens L4 is defined as d7. The total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along the optical axis is defined as TTL. The camera optical lens 10 further satisfies the following condition: $0.04 \leq d7/TTL \leq 0.18$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.07 \leq d7/TTL \leq 0.14$.

(37) In the present embodiment, an object side surface of the fifth lens L5 is concave in the paraxial region, an image side surface of the fifth lens L5 is concave in the paraxial region, and the fifth lens L5 has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the fifth lens L5 can also be arranged as other convex side surface or concave side surface, such as, convex object side surface and convex image side surface and so on. The fifth lens L5 may have a positive refractive power.

(38) A central curvature radius of the object side surface of the fifth lens L5 is defined as R9, and a central curvature radius of the image side surface of the fifth lens L5 is defined as R10. The camera optical lens further satisfies the following condition:

$0.37 \leq (R9+R10)/(R9-R10) \leq 1.42$, which specifies a shape of the fifth lens L5. When the value is within this range, as the development of the ultra-thin and wide-angle lenses, it is beneficial for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, $0.60 \leq (R9+R10)/(R9-R10) \leq 1.14$.

(39) An on-axis thickness of the fifth lens L5 is defined as d9. The total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along the optical axis is defined as TTL. The camera optical lens 10 further satisfies the following condition: $0.01 \leq d9/TTL \leq 0.08$. When the condition is satisfied, it is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.02 \leq d9/TTL \leq 0.07$.

(40) In the present embodiment, an object side surface of the sixth lens L6 is convex in the paraxial region, an image side surface of the sixth lens L6 is concave in the paraxial region, and the sixth lens L6 has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the sixth lens L6 can be arranged as other convex side surface or concave side surface, such as, concave object side surface and convex image side surface and so on. The sixth lens L6 may have positive refractive power.

(41) The focal length of the camera optical lens 10 is defined as f, and a focal length of the sixth lens L6 is defined as f6. The camera optical lens further satisfies the following condition: $-645.45 \leq f6/f \leq 138.30$. It is beneficial for realizing the excellent imaging quality and the lower sensitivity by controlling the refractive power being within reasonable range. Preferably, the following condition shall be satisfied, $-403.41 \leq f6/f \leq 110.64$.

(42) A central curvature radius of the object side surface of the sixth lens L6 is defined as R11, and a central curvature radius of the image side surface of the sixth lens L6 is defined as R12. The camera optical lens further satisfies the following condition:

$0.35 \leq (R11+R12)/(R11-R12) \leq 64.63$, which specifies a shape of the sixth lens L6. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it benefits for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, $0.57 \leq (R11+R12)/(R11-R12) \leq 51.70$.

(43) An on-axis thickness of the sixth lens L6 is defined as d11. The total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along the optical axis is defined as TTL. The camera optical lens further satisfies the following condition: $0.01 \leq d11/TTL \leq 0.12$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.01 \leq d11/TTL \leq 0.09$.

(44) In the present embodiment, an object side surface of the seventh lens L7 is convex in the paraxial region, an image side surface of the seventh lens L7 is convex in the paraxial region, and the seventh lens L7 has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the seventh lens L7 can be arranged as other convex side surface or concave side surface, such as, concave object side surface and concave image side surface and so on. The seventh lens L7 may have a negative refractive power.

(45) The focal length of the camera optical lens 10 is defined as f, and a focal length of the seventh lens L7 is defined as f7. The camera optical lens 10 further satisfies the following condition: $0.40 \leq f7/f \leq 1.59$. It is beneficial for realizing the better imaging quality and the lower sensitivity by controlling the refractive power being within reasonable range. Preferably, the following condition shall be satisfied, $0.64 \leq f7/f \leq 1.27$.

(46) A central curvature radius of the object side surface of the seventh lens L7 is defined as R13, and a central curvature radius of the image side surface of the seventh lens L7 is defined as R14. The camera optical lens 10 further satisfies the following condition:

$0.38 \leq (R13+R14)/(R13-R14) \leq 1.29$, which specifies a shape of the seventh lens L7. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, $0.60 \leq (R13+R14)/(R13-R14) \leq 1.03$.

(47) An on-axis thickness of the seventh lens L7 is defined as d13. The total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along the optical axis is defined as TTL. The camera optical lens further satisfies the following condition: $0.05 \leq d13/TTL \leq 0.31$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.08 \leq d13/TTL \leq 0.25$.

(48) In the present embodiment, the object side surface of the eighth lens L8 is convex in the paraxial region, the image side surface of the eighth lens L8 is concave in the paraxial region, and the eighth lens L8 has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the eighth lens L8 can be arranged as other convex side surface or concave side surface, such as, concave object side surface and convex image side surface and so on. The eighth lens L8 may have a positive refractive power.

(49) The focal length of the camera optical lens 10 is defined as f, and a focal length of the eighth lens L8 is defined as f8. The camera optical lens 10 further satisfies the following condition: $-2.96 \leq f8/f \leq -0.69$. It is beneficial for realizing the better imaging quality and the lower sensitivity by controlling the refractive power being within reasonable range. Preferably, the following condition shall be satisfied, $-1.85 \leq f8/f \leq -0.86$.

(50) A central curvature radius of the object side surface of the eighth lens L8 is defined as R15, and a central curvature radius of the image side surface of the eighth lens L8 is defined as R16. The camera optical lens 10 further satisfies the following condition:

$0.75 \leq (R15+R16)/(R15-R16) \leq 3.80$, which specifies a shape of the eighth lens L8. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, $1.20 \leq (R15+R16)/(R15-R16) \leq 3.04$.

(51) An on-axis thickness of the eighth lens L8 is defined as d15. The total optical length from the object side surface of the first lens L1

to the image surface **S1** of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens further satisfies the following condition: $0.03 \leq d15/TTL \leq 0.15$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.05 \leq d15/TTL \leq 0.12$.

(52) In the present embodiment, an image height of the camera optical lens **10** is defined as IH. The total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along an optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: $TTL/IH \leq 2.91$, thereby achieving the ultra-thin performance. Preferably, the following condition shall be satisfied, $TTL/IH \leq 2.83$.

(53) In the present embodiment, an F number (FNO) of the camera optical lens **10** is smaller than or equal to 2.68, thereby achieving the large aperture and good imaging performance. Preferably, the FNO of the camera optical lens **10** is smaller than or equal to 2.63.

(54) When the above conditions are satisfied, which makes it is possible that the camera optical lens has excellent optical performances, and meanwhile can meet design requirements of large aperture. According the characteristics of the camera optical lens **10**, it is particularly suitable for a mobile camera lens component and a WEB camera lens composed of high pixel CCD, CMOS.

(55) The following examples will be used to describe the camera optical lens **10** of the present invention. The symbols recorded in each example will be described as follows. The focal length, on-axis distance, central curvature radius, on-axis thickness, inflexion point position, and arrest point position are all in units of mm.

(56) TTL: the total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along the optical axis, the unit of TTL is mm.

(57) F number (FNO): a ratio of an effective focal length of the camera optical lens **10** to an entrance pupil diameter (ENPD).

(58) Preferably, inflexion points and/or arrest points can also be arranged on the object side surface and/or image side surface of the lens, so that the demand for high quality imaging can be satisfied, the description below can be referred for specific implementable scheme.

(59) The design information of the camera optical lens **10** in Embodiment 1 of the present invention is shown in the tables 1 and 2.

(60) TABLE-US-00001 TABLE 1 R d nd vd S1 ∞ d0= -3.079 R1 -7.479 d1= 1.306 nd1 1.5444 v1 55.82 R2 1.935 d2= 0.455 R3 2.339 d3= 0.332 nd2 1.6610 v2 20.53 R4 4.207 d4= 0.234 R5 89.494 d5= 0.576 nd3 1.6700 v3 19.39 R6 109.026 d6= 0.138 R7 3.749 d7= 1.151 nd4 1.5444 v4 55.82 R8 -2.873 d8= 0.062 R9 -60.406 d9= 0.219 nd5 1.6700 v5 19.39 R10 8.814 d10= 0.737 R11 13.005 d11= 0.763 nd6 1.6359 v6 23.82 R12 12.415 d12= 0.050 R13 12.224 d13= 1.029 nd7 1.5444 v7 55.82 R14 -1.72 d14= 0.489 R15 9.977 d15= 0.559 nd8 1.6400 v8 23.54 R16 1.997 d16= 1.285 R17 ∞ d17= 0.275 ndg 1.5168 vg 64.20 R18 ∞ d18= 0.110

(61) where, the meaning of the various symbols is as follows. **S1**: aperture; R: curvature radius of an optical surface, a central curvature radius for a lens; R1: central curvature radius of the object side surface of the first lens **L1**; R2: central curvature radius of the image side surface of the first lens **L1**; R3: central curvature radius of the object side surface of the second lens **L2**; R4: central curvature radius of the image side surface of the second lens **L2**; R5: central curvature radius of the object side surface of the third lens **L3**; R6: central curvature radius of the image side surface of the third lens **L3**; R7: central curvature radius of the object side surface of the fourth lens **L4**; R8: central curvature radius of the image side surface of the fourth lens **L4**; R9: central curvature radius of the object side surface of the fifth lens **L5**; R10: central curvature radius of the image side surface of the fifth lens **L5**; R11: central curvature radius of the object side surface of the sixth lens **L6**; R12: central curvature radius of the image side surface of the sixth lens **L6**; R13: central curvature radius of the object side surface of the seventh lens **L7**; R14: central curvature radius of the image side surface of the seventh lens **L7**; R15: central curvature radius of the object side surface of the eighth lens **L8**; R16: central curvature radius of the image side surface of the eighth lens **L8**; R17: central curvature radius of an object side surface of the optical filter GF; R18: curvature radius of an image side surface of the optical filter GF; d: on-axis thickness of a lens and an on-axis distance between lenses; d0: on-axis distance from the aperture **S1** to the object side surface of the first lens **L1**; d1: on-axis thickness of the first lens **L1**; d2: on-axis distance from the image side surface of the first lens **L1** to the object side surface of the second lens **L2**; d3: on-axis thickness of the second lens **L2**; d4: on-axis distance from the image side surface of the second lens **L2** to the object side surface of the third lens **L3**; d5: on-axis thickness of the third lens **L3**; d6: on-axis distance from the image side surface of the third lens **L3** to the object side surface of the fourth lens **L4**; d7: on-axis thickness of the fourth lens **L4**; d8: on-axis distance from the image side surface of the fourth lens **L4** to the object side surface of the fifth lens **L5**; d9: on-axis thickness of the fifth lens **L5**; d10: on-axis distance from the image side surface of the fifth lens **L5** to the object side surface of the sixth lens **L6**; d11: on-axis thickness of the sixth lens **L6**; d12: on-axis distance from the image side surface of the sixth lens **L5** to the object side surface of the seventh lens **L7**; d13: on-axis thickness of the seventh lens **L7**; d14: on-axis distance from the image side surface of the seventh lens **L7** to the object side surface of the eighth lens **L8**; d15: on-axis thickness of the eighth lens **L8**; d16: on-axis distance from the image side surface of the eighth lens **L8** to the object side surface of the optical filter GF; d17: on-axis thickness of the optical filter GF; d18: on-axis distance from the image side surface of the optical filter GF to the image surface; nd: refractive index of d line (d-line is green light with a wavelength of 550 nm); nd1: refractive index of d line of the first lens **L1**; nd2: refractive index of d line of the second lens **L2**; nd3: refractive index of d line of the third lens **L3**; nd4: refractive index of d line of the fourth lens **L4**; nd5: refractive index of d line of the fifth lens **L5**; nd6: refractive index of d line of the sixth lens **L6**; nd7: refractive index of d line of the seventh lens **L7**; nd8: refractive index of d line of the eighth lens **L8**; ndg: refractive index of d line of the optical filter GF; vd: abbe number; v1: abbe number of the first lens **L1**; v2: abbe number of the second lens **L2**; v3: abbe number of the third lens **L3**; v4: abbe number of the fourth lens **L4**; v5: abbe number of the fifth lens **L5**; v6: abbe number of the sixth lens **L6**; v7: abbe number of the seventh lens **L7**; v8: abbe number of the eighth lens **L8**; vg: abbe number of the optical filter GF;

(62) Table 2 shows the aspherical surface data of the camera optical lens **10** in Embodiment 1 of the present invention.

(63) TABLE-US-00002 TABLE 2 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 2.2006E+00 4.3507E-02 -1.2003E-02 2.7849E-03 -4.8370E-04 6.0133E-05 R2 8.3815E-01 8.6747E-02 -2.5639E-02 -2.8534E-04 3.5873E-03 -1.0567E-03 R3 1.2495E+00 -1.9668E-02 -1.3041E-02 1.3000E-03 1.2574E-03 7.4346E-04 R4 -1.9969E+00 -1.6163E-03 6.5094E-03 2.1667E-03 1.9578E-03 1.5277E-03 R5 1.7941E+03 2.7481E-03 -3.8342E-03 -9.0799E-04 2.7356E-04 1.0816E-03 R6 4.7445E+03 1.3601E-04 -1.2405E-02 5.3279E-03 5.7504E-03 4.9519E-03 R7 1.0753E+01 -1.3083E-02 -2.5558E-02 6.0440E-04 -1.7726E-02 -7.6010E-03 R8 4.7219E+00 -1.1832E-01 7.4041E-02 -8.3437E-02 7.2312E-02 -5.5362E-02 R9 2.7091E+03 -2.1282E-01 7.6881E-02 -9.1082E-02 5.8430E-02 -3.5492E-02 R10 -6.0510E+01 -8.7439E-02 3.7036E-02 -1.4442E-02 7.0042E-03 -2.1623E-03 R11 -3.3700E+02 -2.3259E-03 3.1154E-04 8.6345E-06 -6.7870E-06 -1.0480E-06 R12 -1.1425E+02 -6.4922E-03 -1.0148E-03 -1.3697E-04 -6.5904E-06 5.2212E-07 R13 1.8623E+01 -1.3940E-02 -5.0040E-04 6.8796E-04 -2.2121E-04 -1.3589E-05 R14 -2.0315E+00 3.3249E-02 -1.7572E-02 6.1337E-03 -1.3991E-03 2.2819E-04 R15 3.0701E+00 8.9458E-03 -8.4414E-03 1.3469E-03 -9.8952E-05 -3.7634E-06 R16 -4.4084E+00 -6.3795E-03 -6.2535E-04 1.7896E-04 -2.1204E-05 1.4138E-06 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20

R1 2.2006E+00 -5.1432E-06 2.8260E-07 -8.7977E-09 1.1484E-10 R2 8.3815E-01 -6.3920E-04 -3.2825E-04 -1.4503E-04 5.9978E-05 R3 1.2495E+00 1.4804E-04 -6.4878E-05 -8.3018E-05 2.5939E-05 R4 -1.9969E+00 5.9323E-04 2.2354E-05 -1.9602E-04 -1.6884E-04 R5 1.7941E+03 1.6774E-04 -1.5055E-04 -7.7486E-05 1.0732E-04 R6 4.7445E+03 8.2284E-04 -5.2901E-03 -5.5529E-03 6.8382E-03 R7 1.0753E+01 1.0705E-02 -1.1613E-02 1.5403E-02 -4.1957E-02 R8 4.7219E+00 3.0551E-02 -4.7274E-03 -8.8967E-04 -4.0382E-04 R9 2.7091E+03 1.0974E-02 -1.4727E-03 7.8710E-04 1.9829E-04 R10 -6.0510E+01 3.2233E-04 6.7714E-06 2.7774E-05 -1.1130E-05 R11 -3.3700E+02 2.0528E-08 2.6469E-08 4.8590E-09 -7.0569E-10 R12 -1.1425E+02 1.3088E-07 8.5478E-09 -1.3853E-10 -1.1202E-10 R13 1.8623E+01 1.8388E-05 -3.4453E-06 2.1272E-07 -1.8546E-09 R14 -2.0315E+00 -2.5074E-05 1.4677E-06 -3.7363E-08 4.9811E-10 R15 3.0701E+00 -9.7651E-08 4.7247E-08 1.6278E-08 -3.2831E-09 R16 -4.4084E+00 -5.5540E-08 1.0955E-09 -9.2673E-12 1.2493E-13

(64) For convenience, an aspheric surface of each lens surface uses the aspheric surfaces shown in the below condition (1). However, the present invention is not limited to the aspherical polynomials form shown in the condition (1).

$$z=(cr.\text{sup}.2)/\{1+[1-(k+1)$$

$$(c.\text{sup}.2r.\text{sup}.2)].\text{sup}.1/2\}+A4r.\text{sup}.4+A6r.\text{sup}.6+A8r.\text{sup}.8+A10r.\text{sup}.10+A12r.\text{sup}.12+A14r.\text{sup}.14+A16r.\text{sup}.16+A18r.\text{sup}.18+A20r.\text{sup}.20$$

(65) Where, K is a conic coefficient, A4, A6, A8, A10, A12, A14, A16, A18, A20 are aspheric surface coefficients. c is the curvature at the center of the optical surface. r is a vertical distance between a point on an aspherical curve and the optic axis, and z is an aspherical depth (a vertical distance between a point on an aspherical surface, having a distance of r from the optic axis, and a surface tangent to a vertex of the aspherical surface on the optic axis).

(66) Table 3 and Table 4 show design data of inflexion points and arrest points of respective lens in the camera optical lens **10** according to Embodiment 1 of the present invention. P1R1 and P1R2 represent the object side surface and the image side surface of the first lens L1, P2R1 and P2R2 represent the object side surface and the image side surface of the second lens L2, P3R1 and P3R2 represent the object side surface and the image side surface of the third lens L3, P4R1 and P4R2 represent the object side surface and the image side surface of the fourth lens L4, P5R1 and P5R2 represent the object side surface and the image side surface of the fifth lens L5, P6R1 and P6R2 represent the object side surface and the image side surface of the sixth lens L6, and P7R1 and P7R2 represent the object side surface and the image side surface of the seventh lens L7, and P8R1 and P8R2 represent the object side surface and the image side surface of the eighth lens L8. The data in the column named “inflexion point position” refers to vertical distances from inflexion points arranged on each lens surface to the optical axis of the camera optical lens **10**. The data in the column named “arrest point position” refers to vertical distances from arrest points arranged on each lens surface to the optical axis of the camera optical lens **10**.

(67) TABLE-US-00003 TABLE 3 Number of Inflexion Inflexion points point position 1 point position 2 P1R1 2 0.575 2.785 P1R2 0 // P2R1 0 // P2R2 0 // P3R1 2 0.725 0.895 P3R2 2 0.445 0.705 P4R1 1 0.785 / P4R2 0 // P5R1 0 // P5R2 2 0.335 1.225 P6R1 2 0.935 2.305 P6R2 1 0.725 / P7R1 1 0.745 / P7R2 2 1.695 1.985 P8R1 1 1.065 / P8R2 1 1.265 /

(68) TABLE-US-00004 TABLE 4 Number of arrest points Arrest point position 1 P1R1 1 1.115 P1R2 0 / P2R1 0 / P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 1 0.595 P6R1 1 2.065 P6R2 1 1.215 P7R1 1 1.305 P7R2 0 / P8R1 1 1.595 P8R2 1 2.745

(69) FIG. 2 and FIG. 3 respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 656 nm, 588 nm, 546 nm, 486 nm and 436 nm after passing the camera optical lens **10** according to Embodiment 1. FIG. 4 illustrates a field curvature and a distortion of light with a wavelength of 546 nm after passing the camera optical lens **10** according to Embodiment 1, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

(70) Table 13 shows various values of Embodiments 1, 2 and 3 and values corresponding to parameters which are specified in the above conditions.

(71) As shown in Table 13, Embodiment 1 satisfies the above conditions.

(72) In the present embodiment, the entrance pupil diameter (ENPD) of the camera optical lens **10** is 1.029 mm. The image height of 1.0H is 5.150 mm. The FOV is 133.80°. Thus, the camera optical lens **10** satisfies design requirements of wide-angle while the on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

Embodiment 2

(73) Embodiment 2 is basically the same as Embodiment 1, the meaning of its symbols is the same as that of Embodiment 1, in the following, only the differences are listed.

(74) A sixth lens L6 has an image side surface being convex in the paraxial region. A second lens L2 has a negative refractive power, and the sixth lens L6 has a positive refractive power.

(75) FIG. 5 shows a schematic diagram of a structure of a camera optical lens **20** according to Embodiment 2 of the present invention. Table 5 and table 6 show the design data of a camera optical lens **20** in Embodiment 2 of the present invention.

(76) TABLE-US-00005 TABLE 5 R d nd vd S1 ∞ d0= -3.007 R1 -11.509 d1= 0.702 nd1 1.5444 v1 55.82 R2 5.422 d2= 1.196 R3 8.727 d3= 0.154 nd2 1.5844 v2 28.22 R4 5.591 d4= 0.025 R5 4.244 d5= 0.492 nd3 1.6700 v3 19.39 R6 11.250 d6= 0.402 R7 6.783 d7= 0.908 nd4 1.5444 v4 55.82 R8 -3.602 d8= 0.215 R9 -194.997 d9= 0.569 nd5 1.6700 v5 19.39 R10 5.349 d10= 0.368 R11 3032.878 d11= 0.224 nd6 1.6501 v6 21.44 R12 -203.353 d12= 0.151 R13 15.692 d13= 2.158 nd7 1.5444 v7 55.82 R14 -1.508 d14= 0.214 R15 5.023 d15= 1.038 nd8 1.6400 v8 23.54 R16 1.358 d16= 1.070 R17 ∞ d17= 0.275 ndg 1.5168 vg 64.20 R18 ∞ d18= 0.267

(77) Table 6 shows aspherical surface data of each lens of the camera optical lens **20** in Embodiment 2 of the present invention.

(78) TABLE-US-00006 TABLE 6 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 2.9319E+00 4.4693E-02 -1.2190E-02 2.7871E-03 -4.8355E-04 6.0125E-05 R2 1.4672E+00 5.9052E-02 -8.9279E-03 -8.6529E-04 1.3351E-03 -3.6597E-04 R3 -2.2192E+00 4.5232E-03 -3.9953E-03 -8.7220E-04 -2.3449E-04 3.0081E-04 R4 -3.8845E-01 -1.3104E-03 -9.6556E-04 -6.5828E-04 -1.0618E-04 2.5244E-05 R5 -2.9774E+00 -4.8388E-03 -7.4631E-04 2.2041E-04 -8.5287E-06 -2.5444E-05 R6 2.4935E+01 -3.2385E-04 -1.7779E-03 1.4050E-03 -7.5135E-04 1.0958E-03 R7 1.2823E+01 7.5380E-03 -9.8230E-03 -5.7288E-04 -1.7047E-04 -8.2258E-04 R8 5.4528E+00 -5.0199E-02 5.2900E-02 -7.3638E-02 7.5927E-02 -5.8918E-02 R9 -1.2802E-01 7.6469E-02 -8.1703E-02 6.1748E-02 -3.4434E-02 1.0806E-02 R10 1.3619E-01 -7.0012E-02 3.6847E-02 -1.7941E-02 6.7417E-03 -1.9267E-03 R11 -1.0306E+04 1.4461E-03 1.3523E-04 6.1830E-06 -6.6414E-06 -1.3678E-06 R12 -1.3193E-03 -2.5644E-04 -6.1056E-05 -2.4193E-06 8.1661E-07 2.5493E-07 R13 3.9649E+01 -9.7647E-03 3.0673E-04 6.4457E-04 -2.2965E-04 -1.4259E-05 R14 -1.4395E+00 3.3973E-02 -1.8081E-02 6.0170E-03 -1.3941E-03 2.3043E-04 R15 -4.4518E+01 -1.9262E-03 -4.4021E-03 9.4866E-04 -9.3261E-05 3.5013E-06 R16 -4.6732E+00 -5.3363E-03 -5.1815E-04 1.7981E-04 -2.1576E-05 1.4195E-06 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20 R1 2.9319E+00 -5.1403E-06 2.8294E-07 -8.7611E-09 1.1017E-10 R2 1.4672E+00 2.9539E-05 4.8064E-08 4.6258E-08

3.1686E-08 R3 -2.2192E-05 -2.9983E-06 -4.1820E-07 4.5525E-08 R4 -3.8845E-01 2.2003E-05 6.5781E-06
4.7688E-07 4.8645E-08 R5 -2.9774E+00 -3.6008E-06 1.0275E-05 9.5044E-06 4.5146E-06 R6 2.4935E+01 -4.2880E-04
-9.3349E-05 -1.3816E-05 6.4371E-05 R7 1.2823E+01 -1.2892E-03 -1.3751E-03 -9.7902E-04 -3.9296E-04 R8
5.4528E+00 2.6758E-02 -6.5353E-03 5.6999E-04 -1.7779E-04 R9 -1.2802E-01 -2.1566E-03 1.3914E-04 3.2726E-06
0.0000E+00 R10 1.3619E-01 3.6155E-04 -4.0079E-05 1.8384E-06 4.8191E-08 R11 -1.0306E+04 -2.5829E-08 3.2128E-08
9.1544E-09 -1.1585E-10 R12 -1.3193E-03 4.6802E-08 8.4870E-09 1.5591E-09 0.0000E+00 R13 3.9649E+01 1.8293E-05
-3.4656E-06 2.1039E-07 -1.4836E-09 R14 -1.4395E+00 -2.4780E-05 1.4869E-06 -3.8382E-08 -1.0292E-10 R15
-4.4518E+01 7.3994E-08 -8.7602E-09 1.8618E-10 -1.0705E-12 R16 -4.6732E+00 -5.4568E-08 1.1092E-09 -1.0213E-11
5.5522E-14

(79) Table 7 and table 8 show design data of inflexion points and arrest points of respective lens in the camera optical lens **20** according to Embodiment 2 of the present invention.

(80) TABLE-US-00007 TABLE 7 Number of inflexion points Inflexion point position1 P1R1 1 0.435 P1R2 0 / P2R1 1 1.015 P2R2 0 / P3R1 0 / P3R2 0 / P4R1 1 0.855 P4R2 0 / P5R1 0 / P5R2 1 0.585 P6R1 0 / P6R2 1 1.885 P7R1 1 0.915 P7R2 0 / P8R1 1 0.835 P8R2 1 1.195

(81) TABLE-US-00008 TABLE 8 Number of arrest points Arrest point position1 P1R1 1 0.785 P1R2 0 / P2R1 1 1.445 P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 1 1.215 P6R1 0 / P6R2 1 2.095 P7R1 1 1.645 P7R2 0 / P8R1 1 1.525 P8R2 1 3.295

(82) FIG. 6 and FIG. 7 respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 656 nm, 588 nm, 546 nm, 486 nm and 436 nm after passing the camera optical lens **20** according to Embodiment 2. FIG. 8 illustrates a field curvature and a distortion of light with a wavelength of 546 nm after passing the camera optical lens **10** according to Embodiment 2, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

(83) As shown in Table 13, Embodiment 2 satisfies the various conditions.

(84) In the present embodiment, an entrance pupil diameter (ENPD) of the camera optical lens **20** is 1.376 mm. An image height of 1.0H is 4.350 mm. An FOV is 115.80°. Thus, the camera optical lens **20** satisfies design requirements of wide-angle while on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

Embodiment 3

(85) Embodiment 3 is basically the same as Embodiment 1 and involves symbols having the same meanings as Embodiment 1, and only differences therebetween will be described in the following.

(86) A sixth lens L6 has an object side surface being concave in a paraxial region. A second lens L2 has a negative refractive power.

(87) FIG. 9 shows a schematic diagram of a structure of a camera optical lens **30** according to Embodiment 3 of the present invention.

(88) Tables 9 and 10 show design data of a camera optical lens **30** in Embodiment 3 of the present invention.

(89) TABLE-US-00009 TABLE 9 R d nd vd S1 ∞ d0= -2.651 R1 -16.078 d1= 0.542 nd1 1.5444 v1 55.82 R2 8.795 d2= 1.026 R3 20.565 d3= 0.105 nd2 1.5844 v2 28.22 R4 4.709 d4= 0.149 R5 3.709 d5= 0.393 nd3 1.6700 v3 19.39 R6 11.956 d6= 0.399 R7 7.051 d7= 0.839 nd4 1.5444 v4 55.82 R8 -3.737 d8= 0.248 R9 -48.808 d9= 0.534 nd5 1.6700 v5 19.39 R10 5.858 d10= 0.467 R11 -195.438 d11= 0.151 nd6 1.6501 v6 21.44 R12 33.322 d12= 0.134 R13 19.3 d13= 2.024 nd7 1.5444 v7 55.82 R14 -1.474 d14= 0.265 R15 2.594 d15= 0.775 nd8 1.6400 v8 23.54 R16 1.127 d16= 1.271 R17 ∞ d17= 0.275 ndg 1.5168 vg 64.20 R18 ∞ d18= 0.228

(90) Table 10 shows aspherical surface data of each lens of the camera optical lens **30** in Embodiment 3 of the present invention.

(91) TABLE-US-00010 TABLE 10 Conic coefficient Aspheric surface coefficients k A4 A6 A8 A10 A12 R1 -6.3522E+00 4.6150E-02 -1.2198E-02 2.7802E-03 -4.8442E-04 6.0055E-05 R2 -5.6062E-01 5.8391E-02 -9.0126E-03 -8.6214E-04 1.3157E-03 -3.6905E-04 R3 -8.0782E+00 3.8593E-03 -4.0448E-03 -8.4274E-04 -2.0313E-04 3.2121E-04 R4 -7.2734E-03 -3.9581E-04 -9.0565E-04 -8.6067E-04 -1.7313E-04 2.0948E-05 R5 -2.5867E+00 -4.2515E-03 -1.0778E-03 1.1159E-04 -5.1422E-05 -5.8132E-05 R6 5.2424E+00 -1.6705E-03 -1.5913E-03 1.4836E-03 -6.9072E-04 1.1502E-03 R7 7.0023E+00 6.0446E-03 -9.4580E-03 4.8573E-04 6.8203E-04 -8.8034E-04 R8 5.4138E+00 -5.0595E-02 5.2567E-02 -7.3784E-02 7.5735E-02 -5.9099E-02 R9 1.4075E+03 -1.2886E-01 7.6944E-02 -8.0810E-02 6.2090E-02 -3.4368E-02 R10 -5.6014E-01 -7.0292E-02 3.6548E-02 -1.7971E-02 6.7567E-03 -1.9190E-03 R11 7.1278E+03 1.0555E-03 -9.2075E-06 -1.8601E-05 -9.3365E-06 -1.7253E-06 R12 -1.4733E+03 -1.8166E-03 -2.1906E-04 -4.9239E-05 -1.6305E-06 7.3820E-07 R13 3.3420E+01 -1.0587E-02 3.3402E-04 6.3327E-04 -2.3107E-04 -1.3901E-05 R14 -1.3388E+00 3.3186E-02 -1.8141E-02 6.0078E-03 -1.3953E-03 2.3025E-04 R15 -9.3147E+00 -1.0632E-03 -4.3038E-03 9.5622E-04 -9.2907E-05 3.4961E-06 R16 -3.3287E+00 -7.5519E-03 -3.2272E-04 1.7826E-04 -2.1879E-05 1.4181E-06 Conic coefficient Aspheric surface coefficients k A14 A16 A18 A20 R1 -6.3522E+00 -5.1432E-06 2.8314E-07 -8.6985E-09 1.1885E-10 R2 -5.6062E-01 2.9665E-05 1.5567E-07 3.6715E-08 8.2876E-09 R3 -8.0782E+00 -5.0664E-05 2.9316E-07 2.9594E-07 4.6464E-08 R4 -7.2734E-03 2.8467E-05 1.1926E-05 3.3503E-06 7.8458E-07 R5 -2.5867E+00 -3.0504E-05 -9.7648E-06 -4.3155E-06 -4.9686E-06 R6 5.2424E+00 -4.5504E-04 -1.3116E-04 -4.9732E-05 3.7482E-05 R7 7.0023E+00 -2.3674E-03 -2.7953E-03 -1.1301E-03 3.4082E-03 R8 5.4138E+00 2.6711E-02 -6.4455E-03 6.3245E-04 -1.6264E-04 R9 1.4075E+03 1.0844E-02 -2.1109E-03 1.7495E-04 3.1914E-05 R10 -5.6014E-01 3.6380E-04 -3.9592E-05 1.8675E-06 -7.4266E-10 R11 7.1278E+03 -9.3710E-08 2.4944E-08 1.2395E-08 2.9124E-09 R12 -1.4733E+03 2.2762E-07 3.9937E-08 5.5795E-09 2.1506E-10 R13 3.3420E+01 1.8444E-05 -3.4339E-06 2.1514E-07 -9.8398E-10 R14 -1.3388E+00 -2.4806E-05 1.4843E-06 -3.8612E-08 -9.9278E-11 R15 -9.3147E+00 7.1884E-08 -8.9357E-09 1.8013E-10 -5.0302E-13 R16 -3.3287E+00 -5.3961E-08 1.1285E-09 -1.0053E-11 -1.7644E-14

(92) Table 11 and table 12 show Embodiment 3 design data of inflexion points and arrest points of respective lens in the camera optical lens **30** according to Embodiment 3 of the present invention.

(93) TABLE-US-00011 TABLE 11 Number of Inflexion Inflexion inflexion points point position1 point position2 P1R1 1 0.355 / P1R2 0 // P2R1 1 0.865 / P2R2 0 // P3R1 1 1.145 / P3R2 0 // P4R1 0 // P4R2 0 // P5R1 0 // P5R2 1 0.545 / P6R1 2 0.705 1.245 P6R2 2 0.725 1.925 P7R1 1 0.695 / P7R2 0 // P8R1 1 1.045 / P8R2 1 1.205 /

(94) TABLE-US-00012 TABLE 12 Number of arrest points Arrest point position1 P1R1 1 0.625 P1R2 0 / P2R1 1 1.255 P2R2 0 / P3R1 0 / P3R2 0 / P4R1 0 / P4R2 0 / P5R1 0 / P5R2 1 1.085 P6R1 0 / P6R2 1 1.275 P7R1 1 1.275 P7R2 0 / P8R1 1 2.155 P8R2 0 /

(95) FIG. 10 and FIG. 11 respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 656 nm, 588 nm, 546 nm, 486 nm and 436 nm after passing the camera optical lens **30** according to Embodiment 3. FIG. 12 illustrates a field curvature and a distortion of light with a wavelength of 546 nm after passing the camera optical lens **30** according to Embodiment 3, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

(96) Table 13 in the following lists values corresponding to the respective conditions. In the present Embodiment 3 in order to satisfy the

above conditions.

(97) In the present embodiment, an entrance pupil diameter (ENPD) of the camera optical lens **30** is 1.411 mm. An image height of 1.0H is 3.540 mm. An FOV is 100.00°. Thus, the camera optical lens **30** satisfies design requirements of wide-angle while on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

(98) TABLE-US-00013 TABLE 13 Parameters and conditions Embodiment 1 Embodiment 2 Embodiment 3 FOV 133.80° 115.80° 100.00° f1/f -1.00 -2.11 -3.20 R15/R16 5.00 3.70 2.30 f 2.675 3.145 3.225 f1 -2.680 -6.644 -10.319 f2 7.352 -26.898 -10.389 f3 727.994 9.773 7.780 f4 3.170 4.441 4.594 f5 -11.329 -7.668 -7.682 f6 -863.295 289.970 -43.304 f7 2.832 2.632 2.594 f8 -3.957 -3.240 -3.894 FNO 2.60 2.29 2.29 TTL 9.770 10.428 9.825 IH 5.150 4.350 3.540

(99) It is to be understood, however, that even though numerous characteristics and advantages of the present exemplary embodiments have been set forth in the foregoing description, together with details of the structures and functions of the embodiments, the disclosure is illustrative only, and changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms where the appended claims are expressed.

Claims

1. A camera optical lens, comprising, from an object side to an image side in sequence: a first lens having a negative refractive power, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens; the first lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region; wherein the camera optical lens satisfies the following conditions: $100.00^\circ \leq \text{FOV} \leq 135.00^\circ$; $3.20 \leq f_1/f \leq -1.00$; $2.30 \leq R_{15}/R_{16} \leq 5.00$; $0.15 \leq (R_1+R_2)/(R_1-R_2) \leq 0.88$; and $0.03 \leq d_1/\text{TTL} \leq 0.20$; where, FOV: a field of view of the camera optical lens in a diagonal direction; f: a focal length of the camera optical lens; f1: a focal length of the first lens; R1: a central curvature radius of the object side surface of the first lens; R2: a central curvature radius of the image side surface of the first lens; d1: an on-axis thickness of the first lens; TTL: a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis; R15: a central curvature radius of an object side surface of the eighth lens; and R16: a central curvature radius of an image side surface of the eighth lens.
2. The camera optical lens according to claim 1, wherein, the fifth lens has a negative refractive power; wherein the camera optical lens satisfies the following condition: $-4.50 \leq f_5/f \leq -2.00$; f5: a focal length of the fifth lens.
3. The camera optical lens according to claim 1 further satisfying the following conditions: $0.23 \leq (R_1+R_2)/(R_1-R_2) \leq 0.71$; and $0.04 \leq d_1/\text{TTL} \leq 0.16$.
4. The camera optical lens according to claim 1, wherein, the second lens has an object side surface being convex in a paraxial region, an image side surface being concave in the paraxial region; the camera optical lens further satisfies the following conditions: $-17.11 \leq f_2/f \leq 4.12$; $-7.01 \leq (R_3+R_4)/(R_3-R_4) \leq 6.85$; and $0.01 \leq d_3/\text{TTL} \leq 0.05$; where, f2: a focal length of the second lens; R3: a central curvature radius of the object side surface of the second lens; R4: a central curvature radius of the image side surface of the second lens; and d3: an on-axis thickness of the second lens.
5. The camera optical lens according to claim 4 further satisfying the following conditions: $-10.69 \leq f_2/f \leq 3.30$; $-4.38 \leq (R_3+R_4)/(R_3-R_4) \leq 5.48$; and $0.01 \leq d_3/\text{TTL} \leq 0.04$.
6. The camera optical lens according to claim 1, wherein, the third lens has an object side surface being convex in a paraxial region, an image side surface being concave in the paraxial region; the third lens has a positive refractive power; the camera optical lens further satisfies the following conditions: $1.21 \leq f_3/f \leq 408.22$; $-20.33 \leq (R_5+R_6)/(R_5-R_6) \leq -1.27$; and $0.02 \leq d_5/\text{TTL} \leq 0.09$; where, f3: a focal length of the third lens; R5: a central curvature radius of the object side surface of the third lens; R6: a central curvature radius of the image side surface of the third lens; and d5: an on-axis thickness of the third lens.
7. The camera optical lens according to claim 6 further satisfying the following conditions: $1.93 \leq f_3/f \leq 326.58$; $-12.70 \leq (R_5+R_6)/(R_5-R_6) \leq -1.58$; and $0.03 \leq d_5/\text{TTL} \leq 0.07$.
8. The camera optical lens according to claim 1, wherein, the fourth lens has an object side surface being convex in a paraxial region, an image side surface being convex in the paraxial region; the fourth lens has a positive refractive power; the camera optical lens further satisfies the following conditions: $0.59 \leq f_4/f \leq 2.14$; $0.07 \leq (R_7+R_8)/(R_7-R_8) \leq 0.46$; and $0.04 \leq d_7/\text{TTL} \leq 0.18$; where, f4: a focal length of the fourth lens; R7: a central curvature radius of the object side surface of the fourth lens; R8: a central curvature radius of the image side surface of the fourth lens; and d7: an on-axis thickness of the fourth lens.
9. The camera optical lens according to claim 8 further satisfying the following conditions: $0.95 \leq f_4/f \leq 1.71$; $0.11 \leq (R_7+R_8)/(R_7-R_8) \leq 0.37$; and $0.07 \leq d_7/\text{TTL} \leq 0.14$.
10. The camera optical lens according to claim 1, wherein, the fifth lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region; the camera optical lens further satisfies the following conditions: $0.37 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 1.42$; and $0.01 \leq d_9/\text{TTL} \leq 0.08$; where, R9: a central curvature radius of the object side surface of the fifth lens; R10: a central curvature radius of the image side surface of the fifth lens; and d9: an on-axis thickness of the fifth lens.
11. The camera optical lens according to claim 10 further satisfying the following conditions: $0.60 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 1.14$; and $0.02 \leq d_9/\text{TTL} \leq 0.07$.
12. The camera optical lens according to claim 1 further satisfying the following conditions: $-645.45 \leq f_6/f \leq 138.30$; $0.35 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 64.63$; and $0.01 \leq d_{11}/\text{TTL} \leq 0.12$; where, f6: a focal length of the sixth lens; R11: a central curvature radius of an object side surface of the sixth lens; R12: a central curvature radius of an image side surface of the sixth lens; and d11: an on-axis thickness of the sixth lens.
13. The camera optical lens according to claim 12 further satisfying the following conditions: $-403.41 \leq f_6/f \leq 110.64$; $0.57 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 51.70$; and $0.01 \leq d_{11}/\text{TTL} \leq 0.09$.
14. The camera optical lens according to claim 1, wherein, the seventh lens has an object side surface being convex in a paraxial region and an image side surface being convex in a paraxial region; the seventh lens has a positive refractive power; the camera optical lens further satisfies the following conditions: $0.40 \leq f_7/f \leq 1.59$; $0.38 \leq (R_{13}+R_{14})/(R_{13}-R_{14}) \leq 1.29$; and $0.05 \leq d_{13}/\text{TTL} \leq 0.31$; where, f7: a focal length of the seventh lens; R13: a central curvature radius of the object side surface of the seventh lens; R14: a central curvature radius of the image side surface of the seventh lens; and d13: an on-axis thickness of the seventh lens.
15. The camera optical lens according to claim 14 further satisfying the following conditions: $0.64 \leq f_7/f \leq 1.27$; $0.60 \leq (R_{13}+R_{14})/(R_{13}-R_{14}) \leq 1.03$; and $0.08 \leq d_{13}/\text{TTL} \leq 0.25$.

16. The camera optical lens according to claim 1, where, the object side surface of the eighth lens is convex in a paraxial region and the image side surface of the eighth lens is concave in the paraxial region; the eighth lens has a negative refractive power; the camera optical lens further satisfies the following conditions: $-2.96 \leq f_8/f \leq -0.69$; $0.75 \leq (R_{15}+R_{16})/(R_{15}-R_{16}) \leq 3.80$; and $0.03 \leq d_{15}/TTL \leq 0.15$; wherein, f_8 : a focal length of the eighth lens; and d_{15} : an on-axis thickness of the eighth lens.

17. The camera optical lens according to claim 16 further satisfying the following conditions: $-1.85 \leq f_8/f \leq -0.86$;

$1.20 \leq (R_{15}+R_{16})/(R_{15}-R_{16}) \leq 3.04$; and $0.05 \leq d_{15}/TTL \leq 0.12$.

18. The camera optical lens according to claim 1 further satisfying the following condition: $TTL/IH \leq 2.91$; where, IH : an image height of the camera optical lens.

19. The camera optical lens according to claim 1, wherein, an FNO of the camera optical lens is less than or equal to 2.68; where, FNO: a ratio of an effective focal length of the camera optical lens to an entrance pupil diameter.
